ANANDASWARUP VADAPALLI

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SUMMARY

Machine learning researcher with a Ph.D. in Computer Science and Engineering, focused on speech, language, and multimodal AI. Specializes in testing whether generative model outputs are actually correct, using controlled benchmarks, statistical analysis, and human review. Has built neural text-to-speech systems, supervises applied research in LLM evaluation, reward modeling, and model efficiency, and has authored 18 papers, including at Interspeech and ACL.

SKILLS

ML & Generative AI: Transformers, representation learning, sequence modeling, supervised fine-tuning, reward modeling, large audio language models, multimodal LLM evaluation, prompt design.

Evaluation & Efficiency: Controlled benchmarking, human evaluation, error analysis, statistical testing (permutation tests, bootstrap intervals, Holm correction), audio captioning metrics, distillation, quantization (FP16, INT8, INT4), latency and carbon-impact measurement.

Tools & Implementation: Python, PyTorch, Hugging Face Transformers, Hugging Face Jobs, Git, GitHub Actions, Bash, reproducible training and evaluation pipelines.

Research Leadership: Experimental design, technical mentorship, implementation review, academic and industry collaboration, scientific writing.

EXPERIENCE

Innovation Mentor | Part-Time *Apr 2025 - Present*
ResearchSpark Hub | www.researchsparkhub.com/board-of-advisors

- Guide ML and GenAI research across reward modeling, image captioning, multimodal prompting, and audio applications.
- Advise research teams on model selection, experimental controls, evaluation methods, and implementation quality.
- Co-authored three papers accepted to ACDSA 2027 on LLM reward evaluation, image-caption benchmarking, and audience-targeted MRI explanations.

Assistant Professor, Machine Learning Track, MSIT Program *Aug 2020 - Present*
Consortium of Institutions of Higher Learning (CIHL), IIIT Hyderabad Campus

- Design and deliver the Machine Learning track curriculum, connecting ML foundations with modern frameworks and applied research workflows.
- Mentor student cohorts through model development, evaluation, and prototyping in NLP and generative modeling.
- Supervise practicum and co-op projects with academic and industry partners, guiding implementation and reproducible experimentation.

Researcher & Ph.D. Candidate *Aug 2010 - Mar 2025*
Language Technologies Research Center (LTRC), IIIT Hyderabad

- Developed sequence models and learned word representations for language tasks, progressing from recurrent networks to fine-tuned BERT.
- Built neural text-to-speech pipelines and evaluated model behavior through objective benchmarks and listening studies.
- Published at Interspeech, ACL, and SN Computer Science; released research code and contributed to Blizzard Challenge systems.

PROJECTS

Acoustic Correctness of Large Audio Language Model Captions on Clotho

- Evaluated GPT-Audio-1.5, Gemini 3.8 Flash, Qwen2.5-Omni-7B, and Audio Flamingo 3 on all 1,045 Clotho v2.1 evaluation recordings, generating 4,180 captions under one shared prompt.
- Scored captions with 12 reference-based and 2 audio-aware metrics (CLAP similarity, MACE) against a matched human leave-one-out baseline, using Holm-corrected permutation tests.
- Found all four models below the human baseline on every reference-based metric, with Audio Flamingo 3 scoring highest.
- Reviewed 200 captions blind to model, first against references and then by listening; GPT-Audio-1.5, third or fourth on reference-based metrics, was rated accurate on 47 of 50 recordings, and 56 of 64 weak-agreement captions had no major acoustic error.
- Released code and data: github.com/anandaswarup/lalm-acoustic-caption-eval

Model Efficiency: Distillation, Quantization, and Carbon Impact

- Supervised benchmarking of NLLB-200 3.3B and distilled 600M models, including FP16, INT8, and INT4 variants, for English-to-French, Hindi, and Kannada translation.
- Evaluation covered Flores+ translations using BLEU, inference latency, and CodeCarbon estimates of inference emissions.
- The distilled FP32 model showed 71–78% lower latency and 63–65% lower estimated emissions, at a cost of 1.3–3.5 BLEU points.
- In human ratings of accuracy and fluency on translations of 100 conversational sentences, INT4 retained mean scores above 4.3/5 across all three languages. Paper: arxiv.org/abs/2509.23990

Reward Function Evaluation for Language-Conditioned Robot Navigation

- Supervised a comparison of geometric rewards, an LLM judge, and a distilled reward network in four controlled experiments on a shared 2-D gridworld.
- Language instructions and iterative feedback increased judge agreement with a rule-based policy from 78% to 100% across 64 situations.
- Feedback also steered the judge toward an imperfect policy, distinguishing reference agreement from action correctness.
- A reward network trained on 258 LLM judgments achieved 0.058 rubric-score MAE but only 48% action agreement on 48 held-out situations.

LLM-Generated Image Caption Evaluation on MS COCO

- Supervised evaluation of 1,000 LLM-generated captions against 5,000 human references from MS COCO using BLEU, METEOR, and CIDEr.
- Matched leave-one-out comparisons benchmarked model and human captions under identical four-reference conditions.
- Human captions showed higher agreement on BLEU-4 and CIDEr, while METEOR slightly favored model captions; model outputs averaged 2.27 times the human caption length.
- A manual review of 50 captions for omissions, additional detail, and reference mismatches distinguished reference agreement from visual correctness.

Audience-Targeted LLM Explanations of Brain MRI Images

- Supervised evaluation of four audience-specific prompts on 400 brain MRI images across four classes, producing 1,594 valid explanations.
- Assessment covered readability, hallucination, safety, and audience fit, with human and AI ratings compared on 160 shared responses.
- Patient and student prompts produced simpler explanations, alongside only 4% exact human–AI agreement on hallucination ratings.
- Comparing explicit tumor-class mentions against dataset labels showed that audience adaptation did not ensure reliable class identification.

End-to-End Neural Text-to-Speech with Prosody and Phrase Break Modeling

- Built a Tacotron2–WaveRNN synthesis pipeline with dynamic convolutional attention and integrated predicted phrase breaks to control pause placement.
- Fine-tuned BERT for phrase break prediction on LibriTTS, achieving 92.10 F1 versus 88.91 for a BiLSTM baseline.
- Evaluated synthesized stories with 70 listeners; pairwise tests favored speech with predicted phrase breaks over synthesis without phrase break prediction.
- Released TTS and phrasing implementations: github.com/anandaswarup/RNN-TTS and github.com/anandaswarup/phrase_break_prediction

EDUCATION

Doctor of Philosophy (Ph.D.) in Computer Science and Engineering *Mar 2025*
International Institute of Information Technology, Hyderabad (IIIT-H)
Thesis: *Prediction and Incorporation of Phrase Break Models in Text-to-Speech Systems*
Advisor: Dr. Kishore Prahallad

Master of Science (M.S.) in Information Technology *Jul 2010*
International Institute of Information Technology, Hyderabad (IIIT-H)
Specialization: Language Technologies

Bachelor of Engineering (B.E.) in Electrical and Electronics Engineering *May 2005*
Osmania University, Hyderabad

PUBLICATIONS

Anandaswarup Vadapalli. Reference agreement is not acoustic correctness: Evaluating large audio language model captions on Clotho. Submitted to *ERC-LM Workshop at ICASSP 2027*

Anika Datta and **Anandaswarup Vadapalli**. Evaluation of reward functions for robots with natural language instructions. Accepted to *International Conference on Artificial Intelligence, Computer, Data Sciences, and Applications (ACDSA 2027)*

Saurav Karthik Sivalanka and **Anandaswarup Vadapalli**. Evaluating LLM-generated image captions against human references: A study on MS COCO. Accepted to *International Conference on Artificial Intelligence, Computer, Data Sciences, and Applications (ACDSA 2027)*

Saketh Raj Kolar and **Anandaswarup Vadapalli**. Brain MRI explanations using LLM prompting: Effects on readability, safety, and tumor-class naming. Accepted to *International Conference on Artificial Intelligence, Computer, Data Sciences, and Applications (ACDSA 2027)*

Aprameya Deshpande, **Anandaswarup Vadapalli**, and Lavanya Prahallad. Predicting hospital readmission in diabetic patients using machine learning approaches. In Hamid R. Arabnia, Leonidas Deligiannidis, Farzan Shenavarmasouleh, Farid Ghareh Mohammadi, and Soheyla Amirian, editors, *Emerging Trends in Health Informatics, Medical Devices, and Large Language Models*, pages 227–233, Cham, 2025. Springer Nature Switzerland

Dhaathri Vijay and **Anandaswarup Vadapalli**. The hidden costs of translation accuracy: Distillation, quantization, and environmental impact. *CoRR*, abs/2509.23990, 2025

Anandaswarup Vadapalli. An investigation of phrase break prediction in an end-to-end TTS system. *SN Comput. Sci.*, 6(2), 2025

Sivanand Achanta, **Anandaswarup Vadapalli**, and Suryakanth V. Gangashetty. IIITH submission for Blizzard Challenge 2017: A BLSTM based SPSS system using MatNN. In *The Blizzard Challenge 2017, Stockholm, Sweden, August 25, 2017*. ISCA, 2017

Anandaswarup Vadapalli and Suryakanth V. Gangashetty. An investigation of recurrent neural network architectures using word embeddings for phrase break prediction. In Nelson Morgan, editor, *17th Annual Conference*

of the International Speech Communication Association, Interspeech 2016, San Francisco, CA, USA, September 8-12, 2016, pages 2308–2312. ISCA, 2016

Sivanand Achanta, Rambabu Banoth, Ayushi Pandey, **Anandaswarup Vadapalli**, and Suryakanth V. Gangashetty. Contextual representation using recurrent neural network hidden state for statistical parametric speech synthesis. In Alan W. Black, editor, *The 9th ISCA Speech Synthesis Workshop, SSW 2016, Sunnyvale, CA, USA, September 13-15, 2016*, pages 172–177. ISCA, 2016

Kishore Prahallad, **Anandaswarup Vadapalli**, Sai Krishna Rallabandi, Santosh Kesiraju, Hema A. Murthy, T. Nagarajan, Bira Chandra Singh, T. Sajani, K. Sreenivasa Rao, Suryakanth V. Gangashetty, Simon King, Keiichi Tokuda, and Alan W. Black. The Blizzard Challenge 2015. In *The Blizzard Challenge 2015, Berlin, Germany, September 11, 2015*. ISCA, 2015

Sai Krishna Rallabandi, **Anandaswarup Vadapalli**, Sivanand Achanta, and Suryakanth V. Gangashetty. IIIT Hyderabad's submission to the Blizzard Challenge 2015. In *The Blizzard Challenge 2015, Berlin, Germany, September 11, 2015*. ISCA, 2015

Sivanand Achanta, **Anandaswarup Vadapalli**, Sai Krishna R., and Suryakanth V. Gangashetty. Significance of maximum spectral amplitude in sub-bands for spectral envelope estimation and its application to statistical parametric speech synthesis. *CoRR*, abs/1508.00354, 2015

Kishore Prahallad, **Anandaswarup Vadapalli**, Santosh Kesiraju, Hema A. Murthy, Swaran Lata, T. Nagarajan, S. R. Mahadeva Prasanna, Hemant A. Patil, Anil Kumar Sao, Simon King, Alan W. Black, and Keiichi Tokuda. The Blizzard Challenge 2014. In *The Blizzard Challenge 2014, Singapore, Singapore, September 19, 2014*. ISCA, 2014

Anandaswarup Vadapalli and Kishore Prahallad. Learning continuous-valued word representations for phrase break prediction. In Haizhou Li, Helen M. Meng, Bin Ma, Engsiong Chng, and Lei Xie, editors, *15th Annual Conference of the International Speech Communication Association, INTERSPEECH 2014, Singapore, September 14-18, 2014*, pages 41–45. ISCA, 2014

Naresh Kumar Elluru, **Anandaswarup Vadapalli**, Raghavendra Elluru, Hema A. Murthy, and Kishore Prahallad. Is word-to-phone mapping better than phone-phone mapping for handling English words? In *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics, ACL 2013, 4-9 August 2013, Sofia, Bulgaria, Volume 2: Short Papers*, pages 196–200. The Association for Computer Linguistics, 2013

Kishore Prahallad, **Anandaswarup Vadapalli**, Naresh Kumar Elluru, Gautam Mantena, Bhargav Pulugundla, Peri Bhaskararao, Hema A. Murthy, Simon King, Vasilis Karaiskos, and Alan W. Black. The Blizzard Challenge 2013 - Indian language task. In *The Blizzard Challenge 2013, Barcelona, Spain, September 3, 2013*. ISCA, 2013

Anandaswarup Vadapalli, Peri Bhaskararao, and Kishore Prahallad. Significance of word-terminal syllables for prediction of phrase breaks in text-to-speech systems for Indian languages. In *The Eighth ISCA Tutorial and Research Workshop on Speech Synthesis, SSW 2013, Barcelona, Spain, August 31 - September 2, 2013*, pages 189–194. ISCA, 2013